



IILM UNIVERSITY, GREATER NOIDA

Cybersecurity & Web Application Security

InternNova Virtual Internship Program



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Roll No. 25SCS1003002279
B.Tech (CSE) & 3rd Semester
6 Weeks | 01 Aug – 11 Sep 2026

Program Overview

6

WEEKS

01 Aug 2026 – 11 Sep 2026

Fully virtual programme



DOMAIN

Cyber Security



MODE

Virtual internship



HOST ORGANIZATION

InternNova



INSTITUTION

IILM University, Greater Noida, U.P.



PROGRAMME

B.Tech (Computer Science & Engineering)

InternNova & the Cybersecurity Track

The internship was conducted through a specialized Cybersecurity Track that delivers guided learning across foundational offensive and defensive security operations, with an emphasis on industry-aligned fundamentals for engineering students.



Structured Learning

Theoretical modules paired with conceptual assessments to build a solid security foundation.



Hands-on Frameworks

Vulnerability testing frameworks and web application security analysis on authorized targets.



Industry-Aligned

Focus on threat awareness, risk analysis methodologies and core cybersecurity fundamentals.

Why Web Security Assessment Matters



The Challenge

Web applications and network infrastructures face continuous risk from misconfigurations, weak input validation and unpatched vulnerabilities. Many systems stay exposed even as organizations rely on them for public and internal workflows.



The Solution Scope

Theory alone is not enough. The internship focused on analyzing web application structures, mapping attack surfaces, recognizing common vulnerabilities and documenting recommended security practices.

THREATS IN FOCUS

SQL Injection

Cross-Site Scripting

Phishing Attacks

Credential Weakness

What the Internship Set Out to Achieve



Networking Fundamentals

IP & MAC addressing, OSI layer functions and protocol operations.



Threat Intelligence

Threat intelligence principles, social engineering and phishing analysis.



Scanning & Monitoring

Network scanning, service discovery and traffic monitoring basics.



Vulnerability Assessment

Assess authorized test web applications for SQL Injection and XSS.



Risk Scoring & Reporting

Risk scoring methodologies and structured security assessment reports.

Three Learning Domains



Networking & Protocols

IP/MAC structures, routing fundamentals, packet flow and browser-server interaction.



Threat & Phishing Awareness

Email header structures, social engineering techniques and common malware delivery mechanisms.



Web Application Security

Testing designated targets for input validation issues, parameter handling and basic web security flaws.

The Toolkit Behind the Internship



Operating Systems

Linux (Debian / Fedora / Arch) and Windows



Network & Scanning

Nmap, Wireshark, Netcat



Browser Dev Tools

Inspect Element, Network Tab, Storage Inspector



Languages & Scripting

Python, Shell Scripting (Bash)



Documentation

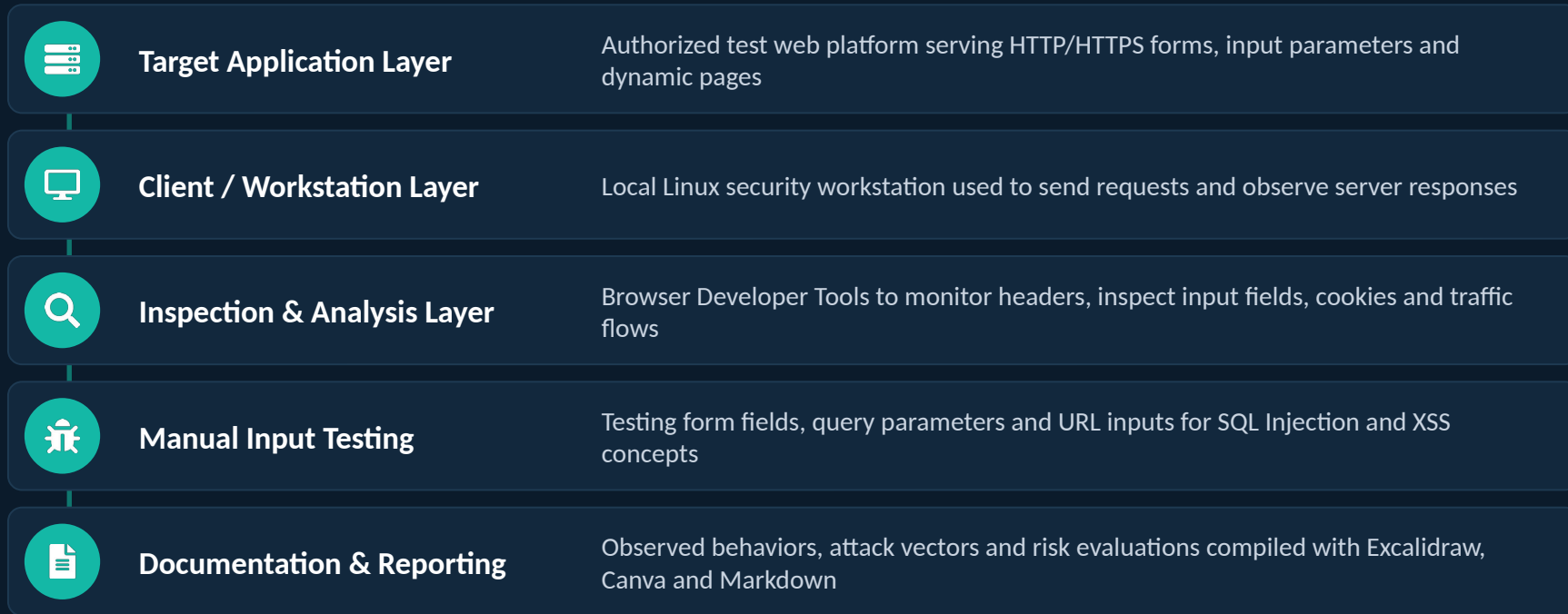
Excalidraw, Canva, Markdown



Assessment Targets

Authorized test web platforms, e.g. testasp.vulnweb.com

Assessment Workflow



The 6-Week Learning Journey

1

WEEK 1

Computer Networking & Protocol Analysis

OSI/TCP-IP models, subnetting, MAC mapping, packet encapsulation, HTTP/HTTPS, DNS and ARP behavior.

2

WEEK 2

Threat Intelligence & Social Engineering

Threat actor tactics, phishing mechanics, SMTP/email header analysis and payload delivery vectors.

3

WEEK 3

Network Reconnaissance & Port Scanning

Active/passive footprinting, host discovery, TCP SYN/Connect scans and Nmap service fingerprinting.

4

WEEK 4

Web Application Vulnerability Assessment

Auditing the test application, parameter handling, validation bypasses and OWASP Top 10 (SQLi, XSS).

5

WEEK 5

Vulnerability Analysis & Risk Quantification

Mapping misconfigurations, classifying severity with risk metrics and detailing remediation.

6

WEEK 6

Security Evaluation & Final Capstone

Python Security Log Analyzer, compiled vulnerability findings and the final technical report.

Capstone & Assessment Output



WEEK 6 CAPSTONE

Security Log Analyzer & Suspicious Activity Detector

A Python-based tool built to analyze security logs and flag suspicious activity.



Web Application Assessment

Structured security evaluation of an authorized test application.



Risk Classification

Findings mapped to severity levels using risk assessment metrics.



Final Technical Report

Vulnerabilities, potential attack vectors and remediation recommendations.

Skills & Capabilities Gained



Networking & Protocols

Clear conceptual understanding of computer networking principles and internet security protocols.



Vulnerability Recognition

Ability to recognize common web application vulnerabilities such as SQL Injection and XSS.



Browser-Based Analysis

Practical familiarity with developer tools for analyzing web responses and form fields.



Security Reporting

Preparing structured assessment reports with observed vulnerabilities and mitigation recommendations.

Successfully Completed



Internship Completion Certificate

Awarded by InternNova in recognition of the successful completion of the 6-week Cyber Security Internship Program (01 Aug – 11 Sep 2026). The certificate is QR-verifiable, and the programme is registered on the AICTE Internship Portal.

CERTIFICATE CRITERIA MET

- ✓ Assigned tasks completed
- ✓ Internship guidelines followed
- ✓ Professional conduct maintained
- ✓ Work submitted within deadlines



Thank You

Cybersecurity & Web Application Security Assessment Internship

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